An Equities Capital Allocation Framework

gappy

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1 Introduction

We consider the problem of allocation risk to multiple investment units ("strategies") controlled by a parent firm. This is a central problem in investment. We list below our assumptions, the main thrust behind the approach, and the findings. We don't claim that this approach is universal. We critique our assumptions, and highlights their limits. However, we show how alternative methods can address them.

1.1 Assumptions

1. The measure of risk employed is dollar volatility, both at the strategy and firm level.
2. The objective of the firm is to maximize dollar profit during the investment period.
3. The firm has a risk budget.
4. Investment units may have additional constraints, such as a maximum risk that they want to deploy.
5. The Sharpe Ratio of the strategies is uncertain. An important component of the framework is the modeling of the uncertainty and its incorporation in the optimization problem.
6. Transaction costs follow the square root law of Kyle.
7. When strategies increase or decrease their risk, they do so without changing their trading behavior.

1.2 Applicability and Limits

1. We use volatility and Sharpe Ratio as measures of risk and performance, respectively. The criticisms levied against the use of volatility as a sufficient risk measure apply here. Loosely speaking, strategies that are based on differences in beliefs about future states of the world are better suited for this approach. These include discretionary and systematic investing in many asset classes, provided that they entail broad portfolios and with sufficiently long holding times. These requirements mitigate the impact of extreme events in the strategy. Long/short and systematic equity, systematic credit and macro are suitable for the framework. Strategies that rely on exploitation of short-horizon market microstructure (high-frequency trading), narrow strategies (macro directional), strategies that rely on market imperfections (index rebalancing and arbitrage strategies like ETF AMP, converts, D1, EFP, etc.) are not suitable for the analysis.

2. The objective function is sensible for most hedge funds. Furthermore the mean-variance formulation is suitable as an approximation of log-growth (Kelly criterion).

3. The square root market impact law holds across orders of magnitude of participation rates, asset classes, and trading mechanisms. However, we ignore propagation effects in the market impact. We believe this assumption to be effectively justified in almost all interesting applications.

4. It is possible to include cross-impact term among strategies, although we do not purse such an extension here. First, this is a second-order correction to transaction costs, and to the overall capital allocation problem. Secondly, estimating the cross-impact is not trivial.

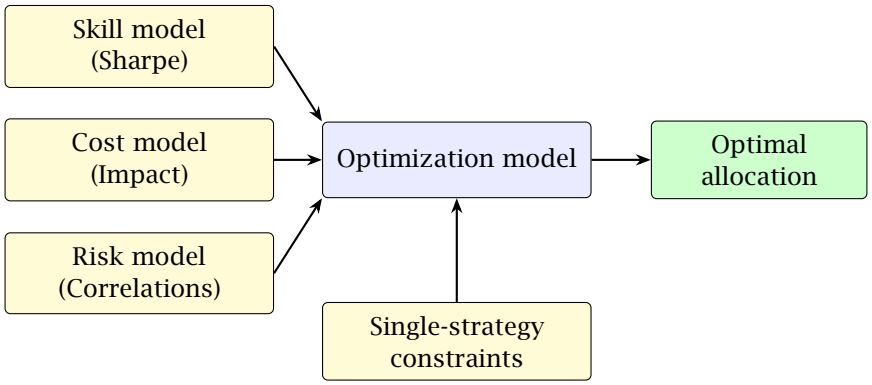
5. Modeling Sharpe distribution of a PM, on the other side, is an essential element of the entire framework. An any point in time, we model the Sharpe Ratio of a PM as a normally distributed random variable with known mean and variance. When hiring a portfolio manager, there is a high degree of uncertainty about their performance, and understanding how it gets resolved over time is important for capital allocation.

6. We assume that the correlations between strategy returns are stable over time. This assumption can be checking empirically. We find that, in general the correlations are stable in applications.

2 Model

2.1 Qualitative formulation

The model consists of three empirical inputs: a skill model, a cost model, and a risk model. Together with single-strategy constraints provided by the firm, they are part of the optimization problem that generates the optimal allocation.



Natural-language description of the capital allocation problem

maximize the sum of the strategies' expected PnL
where each strategy PnL is the expected value of strategy Sharpe × strategy volatility
minus the strategy's execution costs
and the firm's risk is less or equal than a given volatility budget
and each strategy's volatility is greater than zero and possibly bounded above

2.2 Skill Model

For each portfolio manager (PM), the skill model combines their track record with historical PM performance. The calculation has two steps: estimate a prior distribution from historical data, then update it using the PM's observed returns. We update the skill distribution at the PM level[1].

To estimate the prior, we first calculate an empirical annual Sharpe Ratio for each historical PM using their full life-to-date return history. Let $r_{i,t}$ be PM i's daily return on GMV, n_i the number of daily observations, and $T = 250$ the number of trading days per year. With $\overline{r}_i$ and $\hat{\sigma}_{r_i}$ denoting the sample mean and standard deviation of daily returns, the life-to-date Sharpe estimate is

$$\hat{s}_i = \sqrt{T}\,\frac{\overline{r}_i}{\hat{\sigma}_{r_i}}. \tag{1}$$

We then compute the sample mean of these Sharpe estimates within each sector and their sample variance across historical PMs. The sector mean m_k supplies the prior mean, and the pooled variance τ_0^2 supplies the prior variance. This is the *empirical-Bayes* step: the parameters of the prior are estimated from the historical cross-section. For a PM in sector $k(i)$, we use

$$s_i \sim N(m_{k(i)}, \tau_0^2). \tag{2}$$

To update this prior, calculate a current PM's own life-to-date Sharpe $\hat{s}_i$ using the same formula. Under the independent-return approximation, $h_i = n_i/T$ is the length of the track record in years. It enters the update because the annualized Sharpe estimate has approximate sampling variance $T/n_i = 1/h_i$: a one-year record has a Sharpe standard error of approximately one, whereas a four-year record has a standard error of approximately one-half. Thus h_i measures the information in the PM's track record.

Here $\hat{s}_i$ is the same empirical Sharpe estimate defined above; once the returns have been observed, it is a known number. The unknown quantity s_i is the PM's true annual Sharpe Ratio. Different samples of returns would give different values of $\hat{s}_i$, even for the same true s_i. We approximate this sampling error by the normal observation model

$$\hat{s}_i = s_i + \frac{Z_i}{\sqrt{h_i}}, \qquad Z_i \sim N(0,1). \tag{3}$$

The observed estimate is therefore the true Sharpe plus a zero-mean sampling error with standard deviation $1/\sqrt{h_i}$ and variance $1/h_i$. This approximation assumes no serial dependence and an adequately estimated return volatility. If returns are serially dependent, replace n_i by an effective number of independent observations before calculating h_i; h_i then measures the effective length of the record in years.

[1] In BAM, it does so by using an ACU Group for a combined book and the core ACU otherwise. See Capital Allocation Model documentation, `docs/skill.md`, and implementation, `app/skill/bayesian.py`, in `PMDanalytics/capital-allocation-backend`.

The normal prior is conjugate to this normal likelihood, so the updated distribution is also normal. Its parameters are calculated directly. First calculate the weight w_i on the PM's observed Sharpe, then the weighted mean μ_i and posterior variance τ_i^2:

$$w_i := \frac{\tau_0^2}{\tau_0^2 + h_i^{-1}} = \frac{h_i \tau_0^2}{1 + h_i \tau_0^2}, \tag{4}$$

$$\mu_i := (1 - w_i) m_{k(i)} + w_i \hat{s}_i, \tag{5}$$

$$\tau_i^2 := \frac{1}{\tau_0^{-2} + h_i} = \frac{\tau_0^2}{1 + h_i \tau_0^2}. \tag{6}$$

The variance calculation adds the prior's precision τ_0^{-2} to the track record's precision h_i and takes the reciprocal. The resulting posterior is $s_i \mid \hat{s}_i \sim N(\mu_i, \tau_i^2)$.

Thus a short record remains close to the sector prior and has a wide posterior, whereas a longer record receives more weight. We estimate the cross-sectional prior dispersion after excluding PMs with less than nine months of history, and clip sector and PM Sharpe forecasts to predefined ranges. An alternative baseline skill model uses the simpler $w_i = \min\{n_i/780, 1\}$ and an empirically specified, decreasing uncertainty curve. In either case the optimizer receives both μ_i and τ_i; using only the posterior mean would over-allocate to imprecisely measured winners.

Observed returns are net of historical trading costs, while the allocation objective subtracts projected costs at the *proposed* size. Let G_i denote strategy i's gross market value (GMV), G_i^{hist} its average historical GMV, and v_i^{hist} its annualized realized dollar volatility. The function $\widehat{C}_i(G)$ estimates strategy i's annual execution cost in dollars at GMV G. We add the estimated historical annual cost back to net Sharpe to obtain the gross Sharpe input:

$$\mu_i^{\text{gross}} \simeq \mu_i^{\text{net}} + \frac{\widehat{C}_i(G_i^{\text{hist}})}{v_i^{\text{hist}}}. \tag{7}$$

This add-back avoids charging historical costs and projected costs twice. Subsequent formulas use $s_i = \mu_i^{\text{gross}}$ when expected PnL and prospective costs are modeled separately. The diagonal approximation $\mathbf{\Omega_s} = \text{diag}(\tau_1^2, \ldots, \tau_n^2)$ treats uncertainty in PM skill as independent across teams.

2.3 Cost Model

Scaling a PM's book increases both gross opportunity and execution costs. To estimate costs at a proposed book size, we fit a power law using historical monthly GMV, turnover, and trading costs.[2]

For monthly average GMV G_i, monthly turnover $T_i = Q_i/G_i$ (notional traded Q_i divided by GMV), and monthly transaction cost C_i, the fitted relationship is

$$\frac{C_i}{T_i} \simeq a_i G_i^{3/2}, \qquad C_i(G_i) \simeq a_i T_i G_i^{3/2}. \tag{8}$$

The first expression is estimated from monthly observations in a rolling window; sparse or unavailable PM-specific coefficients can be replaced by a pooled estimate. The $3/2$ exponent reflects notional traded growing with size while unit impact grows approximately with its square root. It presumes that turnover and the traded universe remain broadly unchanged under scaling. If monthly turnover and the fitted coefficient persist, the annual cost is $\widehat{C}_i^{\text{yr}}(G_i) \simeq 12 a_i T_i G_i^{3/2}$.

[2] Capital Allocation Model documentation, `docs/market_impact.md`; coefficient fitting in `app/market_impact/market_impact_coefficients.py` and the cost objective in `app/optimizer/opt_setup.py`.

Let $\rho_i > 0$ be annual dollar volatility per dollar of GMV, so that a candidate dollar-volatility allocation x_i requires $G_i = x_i/\rho_i$. Absorbing turnover and annualization into $\tilde{a}_i = 12a_iT_i$, we obtain

$$\hat{C}_i^{\mathrm{yr}}(x_i) = \tilde{a}_i \left(\frac{x_i}{\rho_i}\right)^{3/2}, \tag{9}$$

$$\frac{\partial}{\partial x_i}\left(s_i x_i - \hat{C}_i^{\mathrm{yr}}(x_i)\right) = s_i - \frac{3\tilde{a}_i}{2\rho_i^{3/2}}\sqrt{x_i}. \tag{10}$$

The marginal expression defines the PM's stand-alone efficient scale before portfolio risk, uncertainty, and position constraints. It also shows why transaction costs must be recomputed at each proposed allocation rather than scaled linearly from the observed cost. The allocation objective retains this $3/2$-power cost function. The cost fit omits cross-impact among teams.

2.4 Risk Model

The allocation benefits from diversification only to the extent that strategies' future returns differ. Direct pairwise covariance estimates are particularly noisy for new books. We estimate a sector factor structure and a PM-specific residual component, imputing parameters for short histories.

For each strategy, write its return on GMV as

$$r_{i,t} = \alpha_i + \beta_i F_{k(i),t} + \epsilon_{i,t}, \qquad \mathrm{Cov}(\epsilon_{i,t}, F_{k(i),t}) = 0, \tag{11}$$

where $F_{k(i),t}$ is its sector's average return on GMV. Estimate the cross-sector covariance $\mathbf{\Omega_f}$ from contemporaneous sector returns, and estimate each β_i and residual variance by regression with an intercept. Take $\mathbf{B}$ to have factors in its rows and strategies in its columns, matching the optimizer's exposure convention; the covariance estimator stores its transpose. The covariance matrix is therefore

$$\mathbf{\Omega_r} = \mathbf{B}'\mathbf{\Omega_f}\mathbf{B} + \mathbf{\Omega_\epsilon}, \qquad \mathbf{\Omega_\epsilon} = \mathrm{diag}(\mathrm{Var}(\epsilon_1), \ldots, \mathrm{Var}(\epsilon_n)). \tag{12}$$

The diagonal residual term is an assumption, not an empirical finding of zero residual overlap. Positive covariance between sector factors increases the joint risk of allocating to PMs in those sectors even when their individual forecasts are attractive.

We compute sector means after a GMV filter and estimate annual factor covariance from roughly one year of daily returns. We normally require about three months of PM observations for a regression. For a shorter record, we set the own-sector loading to one and impute the residual variance from its sector median (falling back to a sector variance when needed). This limits spurious diversification credits for a new book. The factor decomposition is estimated from realized returns and is not, by itself, a forecast of changing exposures. The optimizer may alternatively use a separately supplied factor-exposure matrix and factor covariance, which need not coincide with the sector-factor estimates.

To express allocations x_i as *annual dollar volatility*, the return covariance used with them must have unit diagonal. If $\mathbf{\Omega_r}$ is first estimated in annualized return-on-GMV units, let $d_i = \sqrt{[\mathbf{\Omega_r}]_{ii}}$ and $\mathbf{D} = \mathrm{diag}(d_1, \ldots, d_n)$. Then the consistently normalized inputs are $\mathbf{B} \leftarrow \mathbf{B}\mathbf{D}^{-1}$ and $\mathbf{\Omega_\epsilon} \leftarrow \mathbf{D}^{-1}\mathbf{\Omega_\epsilon}\mathbf{D}^{-1}$; the resulting covariance is $\mathbf{D}^{-1}\mathbf{\Omega_r}\mathbf{D}^{-1}$. Firm volatility is consequently

$$\sigma_{\mathrm{firm}}(\mathbf{x}) = \sqrt{\mathbf{x}'\mathbf{\Omega_r}\mathbf{x}} = \sqrt{\mathbf{x}'\mathbf{\Omega_\epsilon}\mathbf{x} + (\mathbf{B}\mathbf{x})'\mathbf{\Omega_f}(\mathbf{B}\mathbf{x})}, \tag{13}$$

where $\mathbf{\Omega_r}$ in the last display denotes the *normalized* matrix. One may cap total volatility or impose distinct residual and factor limits as in the formulation below.

2.5 Quantitative Formulation

The allocation problem has two nonnegative decision variables per PM: x_i, its annualized dollar-volatility target, and g_i, its GMV target. All dollar quantities must use the same units. Let s_i be the expected *gross* annual Sharpe from the skill model, τ_i its posterior standard deviation, and ρ_i the annual return volatility per dollar of GMV. As above, write

$$\mathbf{S} = \operatorname{diag}(\rho_1, \dots, \rho_n), \qquad \mathbf{\Omega_s} = \operatorname{diag}(\tau_1^2, \dots, \tau_n^2), \qquad \mathbf{x} = \mathbf{Sg}. \tag{14}$$

The last equality makes $s_i x_i$ expected gross annual dollar PnL. The coefficient a_i^{opt} below is the market-impact coefficient *after* incorporating turnover, annualization, and the chosen dollar units. It is distinct from the fitted monthly coefficient a_i.

The skill-uncertainty penalty follows from the variance of conditional expected PnL. Consider a single strategy with a fixed dollar-volatility allocation x. Write $\tilde{s}$ for its unknown true gross annual Sharpe, whose posterior mean is s and posterior variance is τ^2. All expectations below use the information available when the allocation is chosen. Conditional on $\tilde{s}$, expected gross annual dollar PnL is $M = x\tilde{s}$. Averaging over the posterior distribution of Sharpe gives

$$E[M] = xE[\tilde{s}] = sx, \tag{15}$$
$$E[M^2] = x^2 E[\tilde{s}^2] = x^2(s^2 + \tau^2), \tag{16}$$
$$\operatorname{Var}(M) = E[M^2] - (E[M])^2 = \tau^2 x^2. \tag{17}$$

Subtracting the squared mean from the second moment gives the centered variance. The skill risk is therefore $\tau^2 x^2$. Doubling the allocation doubles expected PnL and quadruples the variance due to skill uncertainty.

Realized PnL also contains return noise. To separate the two sources of risk, let r be the strategy's annual gross return normalized to unit conditional volatility, so that

$$E[r \mid \tilde{s}] = \tilde{s}, \qquad \operatorname{Var}(r \mid \tilde{s}) = 1. \tag{18}$$

Using these conditional moments, the mean, second moment, and centered variance of realized gross PnL $P = xr$ are

$$E[P] = sx, \tag{19}$$
$$E[P^2] = x^2 E[1 + \tilde{s}^2] = x^2(1 + s^2 + \tau^2), \tag{20}$$
$$\operatorname{Var}(P) = E[P^2] - (E[P])^2 = x^2 + \tau^2 x^2. \tag{21}$$

The first variance term is market-return risk conditional on skill; the second is the additional risk arising from uncertainty about skill.

For several strategies, conditional expected PnL is $M = \sum_i x_i \tilde{s}_i$. With posterior Sharpe covariance $\mathbf{\Omega_s} = \operatorname{diag}(\tau_1^2, \dots, \tau_n^2)$, we obtain

$$E[M] = \sum_i s_i x_i, \tag{22}$$
$$\operatorname{Var}(M) = \mathbf{x}'\mathbf{\Omega_s}\mathbf{x} = \sum_i \tau_i^2 x_i^2. \tag{23}$$

We control market-return risk through the volatility constraints and charge λ_{skill} for each unit of uncertainty variance in expected PnL. Subtracting this charge from expected net PnL gives the quadratic term $-\lambda_{\text{skill}}\mathbf{x}'\mathbf{\Omega_s}\mathbf{x}$. The posterior determines the variance; we choose λ_{skill} to specify

how much the firm pays to reduce this source of risk. Projected execution costs are treated as fixed at each proposed allocation and deducted from expected PnL.

With V_{tot} denoting the risk budget and $\lambda_{\text{tc}}, \lambda_{\text{skill}} \geq 0$ denoting cost and skill-uncertainty scalars, the documented total-risk formulation is

$$\max_{\mathbf{x},\mathbf{g}} \quad \sum_{i=1}^{n} s_i x_i - \lambda_{\text{tc}} \sum_{i=1}^{n} a_i^{\text{opt}} g_i^{3/2} - \lambda_{\text{skill}} \mathbf{x}' \boldsymbol{\Omega}_{\mathbf{s}} \mathbf{x} \tag{24}$$

$$\text{s.t.} \quad \mathbf{x} = \mathbf{S}\mathbf{g}, \qquad \mathbf{g} \geq 0, \tag{25}$$

$$\underline{x}_i \leq x_i \leq \overline{x}_i, \qquad i = 1, \ldots, n, \tag{26}$$

$$\sqrt{\mathbf{x}' \boldsymbol{\Omega}_{\mathbf{r}} \mathbf{x}} \leq V_{\text{tot}}. \tag{27}$$

The first term is expected gross PnL, the second is projected transaction cost at the proposed GMV, and the third prices the posterior variance of conditional expected PnL derived above. The coefficient λ_{skill} has inverse-dollar units so that its product with a dollar-PnL variance remains in dollars. Diversification enters through the covariance constraint, rather than an additional correlation penalty in the objective.

Sector and region limits bound their shares of the *sum of individual* dollar-volatility targets, which generally differs from portfolio volatility. Let $\mathcal{S}_k$ and $\mathcal{R}_\ell$ denote the PMs in sector k and region ℓ, respectively. The implemented share bounds take the form

$$\underline{q}_k^{\text{sec}} \sum_i x_i \leq \sum_{i \in \mathcal{S}_k} x_i \leq \overline{q}_k^{\text{sec}} \sum_i x_i, \qquad \text{for each sector } k, \tag{28}$$

$$\underline{q}_\ell^{\text{reg}} \sum_i x_i \leq \sum_{i \in \mathcal{R}_\ell} x_i \leq \overline{q}_\ell^{\text{reg}} \sum_i x_i, \qquad \text{for each region } \ell. \tag{29}$$

The bounds $\underline{x}_i, \overline{x}_i$ limit the change in a PM's target relative to its existing allocation. The fractions for each partition must be jointly feasible with those individual bounds.

An alternative to the total-risk cap (27) is to impose separate idiosyncratic and factor-volatility caps:

$$\sqrt{\mathbf{x}' \boldsymbol{\Omega}_{\boldsymbol{\epsilon}} \mathbf{x}} \leq V_\epsilon, \tag{30}$$

$$\mathbf{y} = \mathbf{B}\mathbf{x}, \qquad \sqrt{\mathbf{y}' \boldsymbol{\Omega}_{\mathbf{f}} \mathbf{y}} \leq V_f. \tag{31}$$

Here $\mathbf{B}$ and $\boldsymbol{\Omega}_{\boldsymbol{\epsilon}}$ have the normalized, dollar-volatility convention established in the risk section. If the covariance inputs are instead annual return-on-GMV covariances, the equivalent risk calculation uses $\mathbf{g}$ in these quadratic forms and the *unnormalized* factor exposures; mixing the two conventions would misstate the risk budget. An aggregate cap in addition to the two component caps would be a different, potentially more restrictive specification.

The $g_i^{3/2}$ penalty is convex for $g_i \geq 0$. With nonnegative penalty coefficients and positive-semidefinite covariances, the objective is concave and the feasible set is convex. The next section retains the $3/2$-power cost function, solves the diagonal-covariance problem exactly, and approximates the effect of the off-diagonal covariance.

3 Simplifications and Approximations

We now retain only the firm's aggregate risk budget and nonnegative allocations. There are no PM allocation bounds or sector and region share limits. The execution-cost and skill-uncertainty

terms remain as specified above. Eliminating GMV through $g_i = x_i/\rho_i$, define

$$c_i := \frac{\lambda_{\text{tc}} a_i^{\text{opt}}}{\rho_i^{3/2}}, \qquad q_i := \lambda_{\text{skill}} \tau_i^2, \qquad V := V_{\text{tot}} > 0. \tag{32}$$

Take $c_i, q_i \geq 0$. The objective is therefore $f(\mathbf{x}) = \sum_i (s_i x_i - c_i x_i^{3/2} - q_i x_i^2)$.

Separate the return covariance into its diagonal and off-diagonal parts:

$$\mathbf{\Omega}_{\mathbf{r}}(\eta) = \mathbf{\Omega}_d + \eta \mathbf{\Omega}_o, \qquad \mathbf{\Omega}_d = \text{diag}(\omega_1, \ldots, \omega_n), \qquad \omega_i > 0. \tag{33}$$

The matrix $\mathbf{\Omega}_o$ is symmetric with zero diagonal. The parameter η connects the diagonal problem at $\eta = 0$ to the full covariance at $\eta = 1$. Since returns are normalized to unit volatility, normally $\omega_i = 1$ and $\mathbf{\Omega}_d = \mathbf{I}$. We assume

$$\left\| \mathbf{\Omega}_d^{-1/2} \mathbf{\Omega}_o \mathbf{\Omega}_d^{-1/2} \right\|_2 \ll 1. \tag{34}$$

This condition controls the combined effect of the off-diagonal entries; small pairwise correlations alone need not imply a small matrix norm. For each η, the allocation problem is

$$\max_{\mathbf{x} \geq 0} \quad f(\mathbf{x}) = \sum_i (s_i x_i - c_i x_i^{3/2} - q_i x_i^2) \tag{35}$$

$$\text{s.t.} \quad \mathbf{x}' \mathbf{\Omega}_{\mathbf{r}}(\eta) \mathbf{x} \leq V^2. \tag{36}$$

Thus the approximation changes the covariance while retaining the $3/2$-power cost function.

3.1 Solve the Diagonal Problem

Let $\nu \geq 0$ be the multiplier of the variance constraint. At $\eta = 0$, the Lagrangian separates into one problem per PM:

$$\max_{x_i \geq 0} \quad s_i x_i - c_i x_i^{3/2} - (q_i + \nu \omega_i) x_i^2. \tag{37}$$

For $s_i > 0$, its first-order condition is

$$s_i - \frac{3}{2} c_i \sqrt{x_i} - 2(q_i + \nu \omega_i) x_i = 0. \tag{38}$$

Setting $z_i = \sqrt{x_i}$ gives a quadratic equation, $2(q_i + \nu \omega_i) z_i^2 + \frac{3}{2} c_i z_i - s_i = 0$. Its positive root yields the allocation

$$\boxed{x_i(\nu) = \left[\frac{2 s_i}{\frac{3}{2} c_i + \sqrt{\frac{9}{4} c_i^2 + 8(q_i + \nu \omega_i) s_i}} \right]^2, \qquad s_i > 0.} \tag{39}$$

For $s_i \leq 0$, set $x_i(\nu) = 0$. The formula above also covers the case $q_i + \nu \omega_i = 0$ when $c_i > 0$. If both cost and skill penalties vanish for a PM with positive Sharpe, its unconstrained allocation is unbounded, so the firm risk budget must bind.

If the unconstrained allocations $x_i(0)$ satisfy $\sum_i \omega_i x_i(0)^2 \leq V^2$, take $\nu_0 = 0$. Otherwise, determine $\nu_0 > 0$ from the scalar equation

$$\sum_i \omega_i x_i(\nu_0)^2 = V^2. \tag{40}$$

In either case, set $x_{0,i} := x_i(\nu_0)$. The left-hand side decreases with ν, so a bracketed scalar root search finds the unique positive multiplier. This solves the diagonal allocation problem exactly. In the zero-cost, zero-skill-penalty limit, $x_{0,i} = s_i/(2\nu_0 \omega_i)$ for positive-Sharpe PMs.

3.2 Add the Off-Diagonal Correction

Assume the diagonal risk budget binds with $\nu_0 > 0$, and that a small covariance perturbation leaves the set of positive allocations unchanged. Let $\mathscr{I} = \{i : x_{0,i} > 0\}$. In the following calculation, restrict vectors and matrices to $\mathscr{I}$; allocations outside this set remain zero. Expand the optimal allocation and multiplier as

$$\mathbf{x}^\star(\eta) = \mathbf{x}_0 + \eta\,\delta\mathbf{x} + O(\eta^2), \tag{41}$$

$$\nu^\star(\eta) = \nu_0 + \eta\,\delta\nu + O(\eta^2). \tag{42}$$

The first-order condition for an active PM in the full problem is

$$s_i - \frac{3}{2}c_i\sqrt{x_i} - 2q_i x_i - 2\nu(\mathbf{\Omega}_\mathbf{r}(\eta)\mathbf{x})_i = 0. \tag{43}$$

Define the positive diagonal curvature matrix $\mathbf{H}$ and the vector $\mathbf{u}$:

$$\mathbf{H} := \operatorname{diag}_{i\in\mathscr{I}}\left(\frac{3c_i}{4\sqrt{x_{0,i}}} + 2q_i + 2\nu_0\omega_i\right), \qquad \mathbf{u} := \mathbf{\Omega}_d\mathbf{x}_0. \tag{44}$$

Differentiating the first-order conditions and the binding risk constraint at $\eta = 0$ gives

$$\mathbf{H}\,\delta\mathbf{x} + 2\mathbf{u}\,\delta\nu = -2\nu_0\mathbf{\Omega}_o\mathbf{x}_0, \tag{45}$$

$$2\mathbf{u}'\delta\mathbf{x} + \mathbf{x}_0'\mathbf{\Omega}_o\mathbf{x}_0 = 0. \tag{46}$$

Solving these equations yields

$$\delta\nu = \frac{\frac{1}{4}\mathbf{x}_0'\mathbf{\Omega}_o\mathbf{x}_0 - \nu_0\mathbf{u}'\mathbf{H}^{-1}\mathbf{\Omega}_o\mathbf{x}_0}{\mathbf{u}'\mathbf{H}^{-1}\mathbf{u}}, \tag{47}$$

$$\delta\mathbf{x} = -2\mathbf{H}^{-1}\left(\nu_0\mathbf{\Omega}_o\mathbf{x}_0 + \delta\nu\,\mathbf{u}\right). \tag{48}$$

The term $\mathbf{\Omega}_o\mathbf{x}_0$ measures each PM's covariance overlap with the baseline allocation. At a fixed multiplier, positive overlap reduces its allocation. The adjustment $\delta\nu$ redistributes risk across the firm so that the budget remains satisfied to first order. Holding the multiplier fixed would generally violate that condition.

At $\eta = 1$, use $\widetilde{\mathbf{x}} = \mathbf{x}_0 + \delta\mathbf{x}$ as the first-order approximation. Reinsert zero allocations for PMs outside $\mathscr{I}$ and calculate risk using the full covariance. A common rescaling puts the approximation exactly on the binding risk budget:

$$\hat{\mathbf{x}} = \frac{V\widetilde{\mathbf{x}}}{\sqrt{\widetilde{\mathbf{x}}'\mathbf{\Omega}_\mathbf{r}\widetilde{\mathbf{x}}}}. \tag{49}$$

Under the stated regularity conditions, the risk discrepancy and the rescaling are second order in the off-diagonal covariance. The approximation requires positive allocations to remain positive and the active set to stay unchanged. If a PM enters or leaves the portfolio, re-solve the full problem. If the diagonal budget is strictly slack, the unconstrained allocation remains optimal while it is feasible under the full covariance; no covariance correction is then needed.

Finally, let $F(\eta)$ be the optimal objective value in (36). The envelope theorem gives

$$F(\eta) = F(0) - \eta\nu_0\mathbf{x}_0'\mathbf{\Omega}_o\mathbf{x}_0 + O(\eta^2). \tag{50}$$

Positive aggregate covariance overlap lowers the attainable objective; negative overlap provides a diversification benefit.

4 Computations

The full allocation problem can be formulated exactly as a second-order cone program (SOCP), including the $3/2$-power execution costs. This gives us a way to compute the optimum with the full covariance and to check the diagonal approximation. We do not need to construct the cones by hand: CVXPY, introduced by Diamond and Boyd, translates the objective and constraints into a form accepted by a numerical solver.[3]

4.1 Second-Order Cone Representation

For a positive-semidefinite covariance, choose a square-root factor L such that $L'L = \mathbf{\Omega_r}$. The total-risk constraint is then $\|L\mathbf{x}\|_2 \le V_{\text{tot}}$, a second-order cone constraint. An eigenvalue decomposition gives such a factor even when the covariance is singular.

For the cost term, introduce $t_i, u_i \ge 0$ and impose

$$u_i^2 \le g_i, \qquad g_i^2 \le u_i t_i, \qquad g_i \ge 0. \tag{51}$$

These inequalities hold for some u_i exactly when $t_i \ge g_i^{3/2}$: the first gives $u_i \le \sqrt{g_i}$, and the second requires $t_i \ge g_i^2/u_i$ when $g_i > 0$. Conversely, choose $u_i = \sqrt{g_i}$. The case $g_i = 0$ permits $u_i = 0$ and any $t_i \ge 0$. Both inequalities are rotated second-order cone constraints, since

$$z^2 \le ab, \quad a, b \ge 0 \quad \Longleftrightarrow \quad \left\| \begin{pmatrix} 2z \\ a - b \end{pmatrix} \right\|_2 \le a + b. \tag{52}$$

The epigraph of the quadratic skill penalty is also representable by a rotated second-order cone. With these auxiliary variables, the objective is linear and all remaining allocation and share constraints are affine. Thus an SOCP suffices; the original power-law cost need not be approximated.

4.2 A CVXPY Formulation

The following function takes the gross Sharpe vector $\mathbf{s}$, posterior standard deviations $\boldsymbol{\tau}$, annual volatility per dollar of GMV $\boldsymbol{\rho}$, cost coefficients $\mathbf{a}^{\text{opt}}$, normalized return covariance $\mathbf{\Omega_r}$, risk budget, and the two penalty weights. It returns the problem and its dollar-volatility and GMV variables. The expressions `power`, `sum_squares`, and `norm` directly encode the cost, skill variance, and portfolio volatility; CVXPY supplies the auxiliary variables and cone constraints.

By default there are no individual PM bounds or sector and region limits, as in the preceding section. Optional `lower` and `upper` arguments impose PM bounds. Each optional group is a tuple consisting of a zero–one membership vector and its lower and upper shares of $\sum_i x_i$. Sector and region groups can be supplied together. Scalar vector inputs are broadcast to all PMs.

All dollar quantities must use the same units. If GMV, dollar volatility, and annual cost are measured in millions, while the fitted monthly a_i was estimated using dollar GMV and dollar cost, supply

$$a_i^{\text{opt}} = 10^3 \tilde{a}_i = 12 \cdot 10^3 a_i T_i. \tag{53}$$

The factor 10^3 follows from dividing dollar cost by 10^6 while raising GMV's 10^6 conversion to the power $3/2$. Use inverse millions for λ_{skill} in this convention.

[3] Steven Diamond and Stephen Boyd, *CVXPY: A Python-Embedded Modeling Language for Convex Optimization*, JMLR 17(83), 2016; paper and CVXPY documentation.

```
import cvxpy as cp
import numpy as np


def capital_allocation_problem(
    s, tau, rho, a_opt, omega_r, risk_budget,
    lambda_tc=1.0, lambda_skill=1.0,
    lower=None, upper=None, groups=(),
):
    """Return (problem, x, g); call problem.solve() separately.

    All dollar quantities use the same units. omega_r is the
    covariance of returns normalized to unit individual volatility.
    It must be positive definite with condition number <= 1e4.
    Each group is (indicator_vector, lower_share, upper_share).
    """
    s = np.asarray(s, dtype=float)
    if s.ndim != 1 or s.size == 0 or not np.isfinite(s).all():
        raise ValueError("s must be a finite, nonempty vector")
    n = s.size

    def vector(value):
        value = np.broadcast_to(np.asarray(value, float), (n,))
        if not np.isfinite(value).all():
            raise ValueError("vector inputs must be finite")
        return value

    tau, rho, a_opt = map(vector, (tau, rho, a_opt))
    if np.any(tau < 0) or np.any(rho <= 0) or np.any(a_opt < 0):
        raise ValueError("require tau, a_opt >= 0 and rho > 0")
    scalars = np.asarray([risk_budget, lambda_tc, lambda_skill])
    if not np.isfinite(scalars).all() or np.any(scalars < 0):
        raise ValueError("budget and penalty weights must be >= 0")

    omega_r = np.asarray(omega_r, float)
    if omega_r.shape != (n, n) or not np.isfinite(omega_r).all():
        raise ValueError("omega_r must be a finite n-by-n matrix")
    if not np.allclose(omega_r, omega_r.T, rtol=0, atol=1e-10):
        raise ValueError("omega_r must be symmetric")
    eigenvalues, U = np.linalg.eigh((omega_r + omega_r.T) / 2)
    if eigenvalues[0] <= 0:
        raise ValueError("omega_r must be positive definite")
    condition_number = eigenvalues[-1] / eigenvalues[0]
    if condition_number > 1e4:
        raise ValueError("omega_r condition number exceeds 1e4")
    # L.T @ L = omega_r; the covariance is not regularized.
    L = np.sqrt(eigenvalues)[:, None] * U.T

    x = cp.Variable(n, nonneg=True, name="dollar_volatility")
    g = cp.Variable(n, nonneg=True, name="gmv")
    costs = a_opt @ cp.power(g, 1.5)
    skill_variance = cp.sum_squares(cp.multiply(tau, x))
    objective = cp.Maximize(
        s @ x - lambda_tc * costs - lambda_skill * skill_variance
    )
    constraints = [
        x == cp.multiply(rho, g),
        cp.norm(L @ x, 2) <= risk_budget,
```

```
    ]
    if lower is not None:
        constraints.append(x >= vector(lower))
    if upper is not None:
        constraints.append(x <= vector(upper))
    for indicator, share_lower, share_upper in groups:
        indicator = vector(indicator)
        if not np.isin(indicator, [0, 1]).all():
            raise ValueError("group indicators must contain 0 or 1")
        if not 0 <= share_lower <= share_upper <= 1:
            raise ValueError("require 0 <= lower share <= upper <= 1")
        group_risk = indicator @ x
        constraints += [
            group_risk >= share_lower * cp.sum(x),
            group_risk <= share_upper * cp.sum(x),
        ]
    return cp.Problem(objective, constraints), x, g
```

The function requires a positive-definite covariance with spectral condition number

$$\kappa_2(\mathbf{\Omega_r}) = \frac{\lambda_{\max}(\mathbf{\Omega_r})}{\lambda_{\min}(\mathbf{\Omega_r})} \le 10^4. \tag{54}$$

It raises an error for a nonpositive eigenvalue or a condition number above 10^4, before constructing the optimization problem. It does not clip eigenvalues or regularize a rejected covariance. The solve status must be checked before using the allocations; a status of `optimal_inaccurate` also calls for checking the constraint residuals against the required numerical tolerance.

To impose the alternative residual and factor budgets, replace the aggregate norm constraint by $\|L_\epsilon \mathbf{x}\|_2 \le V_\epsilon$ and $\|L_f \mathbf{B}\mathbf{x}\|_2 \le V_f$, with $L_\epsilon' L_\epsilon = \mathbf{\Omega_\epsilon}$ and $L_f' L_f = \mathbf{\Omega_f}$. In the capital-allocation implementation, `app/optimizer/opt_setup.py` uses these two component caps when a separate factor model is supplied; it does not add the documented aggregate cap. Without that factor model, the same file applies the supplied full covariance under the idiosyncratic-risk-limit setting.

4.3 A Ten-PM Example

Draw ten posterior mean gross Sharpes and ten posterior variances, independently across PMs and between the two sets of inputs:

$$s_i \sim U[0,1], \qquad \log(\tau_i^2) \sim N(0.5, 0.2^2), \qquad i = 1, \dots, 10. \tag{55}$$

Thus $\tau_i^2 \sim \text{LOGN}(0.5, 0.2)$ uses 0.5 and 0.2 as the mean and standard deviation of the logarithm, respectively. The function takes posterior standard deviations, so we pass the square root of each sampled variance. Set every cross-PM return correlation to 0.1:

$$\mathbf{\Omega_r} = 0.9\mathbf{I}_{10} + 0.1\mathbf{1}\mathbf{1}'. \tag{56}$$

Here $\mathbf{1}$ is the ten-dimensional vector of ones. The eigenvalues are 0.9 with multiplicity nine and 1.9 with multiplicity one, giving a condition number $1.9/0.9 \simeq 2.11$.

Measure GMV, dollar volatility, and annual PnL in millions. Set the firm risk budget to $V_{\text{tot}} = 500$ and use $\rho_i = 0.2$ for every PM. Annual turnover is $\theta = 10$, defined as total notional traded during the year, counting both purchases and sales, divided by GMV.

To calibrate the common execution-cost curve, assume a cost of $k_{\text{ref}} = 10$ basis points per dollar traded at reference GMV $g_{\text{ref}} = 1000$ ($1 billion). This is an illustrative equity cost benchmark.[4] Let cost per dollar traded grow with the square root of GMV:

$$k(g) = 10^{-3}\sqrt{\frac{g}{1000}}, \qquad C_i(g_i) = \theta g_i k(g_i) = a^{\text{opt}} g_i^{3/2}, \tag{57}$$

$$a^{\text{opt}} = \frac{10 \cdot 10^{-3}}{\sqrt{1000}} \simeq 3.1623 \times 10^{-4}. \tag{58}$$

Thus a $1 billion GMV book trades $10 billion annually and incurs $10 million of annual execution cost, or 1% of GMV. Turnover and the cost-per-trade assumption are already incorporated in a^{opt}. The same function applies to every PM.

Take $\lambda_{\text{tc}} = 1$ and set the dimensionless skill-risk aversion $\lambda_{\text{skill}} V_{\text{tot}} = 0.25$, giving $\lambda_{\text{skill}} = 0.0005$ per million. There are no PM bounds or sector and region share constraints. The fixed random seed makes the example reproducible. After installing `cvxpy` and `numpy`, run the following code with `capital_allocation.py` in the same directory:

```
import cvxpy as cp
import numpy as np
from capital_allocation import capital_allocation_problem

rng = np.random.default_rng(42)
n = 10
s = rng.uniform(low=0.0, high=1.0, size=n)
posterior_variance = rng.lognormal(mean=0.5, sigma=0.2, size=n)
tau = np.sqrt(posterior_variance)
omega_r = 0.9 * np.eye(n) + 0.1 * np.ones((n, n))

# GMV, dollar volatility, and PnL are in millions.
rho = np.full(n, 0.2)
risk_budget = 500.0
annual_turnover = 10.0  # Total annual traded notional / GMV.
reference_gmv = 1000.0
reference_cost_bps = 10.0  # Cost per dollar traded at reference GMV.
cost_rate = reference_cost_bps * 1e-4
cost_coefficient = annual_turnover * cost_rate / np.sqrt(reference_gmv)
a_opt = np.full(n, cost_coefficient)
lambda_tc = 1.0
lambda_skill = 0.25 / risk_budget

problem, x, g = capital_allocation_problem(
    s, tau, rho, a_opt, omega_r, risk_budget,
    lambda_tc=lambda_tc, lambda_skill=lambda_skill,
)
assert problem.is_dcp()
problem.solve()
if problem.status not in (cp.OPTIMAL, cp.OPTIMAL_INACCURATE):
    raise RuntimeError(problem.status)
x_star, g_star = x.value, g.value
portfolio_volatility = np.sqrt(x_star @ omega_r @ x_star)
if portfolio_volatility > risk_budget * (1 + 1e-6):
    raise RuntimeError("risk budget violated")
```

[4] Israel, Moskowitz, Ross, and Serban, *Putting an Academic Factor into Practice: The Case of Momentum* (2021), Table 4, report realized costs per dollar traded of 6.40 bp for U.S. large-cap equities and 18.27 bp for U.S. small-cap equities over 2009–2016. The 10 bp cost and $1 billion reference GMV here are calibration assumptions for the example.

```
print("PM    Sharpe   Variance       x       GMV")
for i in range(n):
    print(f"{i+1:2d}  {s[i]:8.4f}  {posterior_variance[i]:8.4f}"
          f"  {x_star[i]:7.4f}  {g_star[i]:8.4f}")
print(f"Portfolio volatility: {portfolio_volatility:.4f}")
print(f"Objective: {problem.value:.4f}")
```

The resulting allocations are shown below; x_i and g_i are in millions.

PM	s_i	τ_i^2	$x_i^\star$	$g_i^\star$
1	0.7740	1.9658	134.2661	671.3307
2	0.4389	1.9262	51.8780	259.3898
3	0.8586	1.6706	168.0268	840.1340
4	0.6974	2.0657	112.3087	561.5434
5	0.0942	1.8103	0.0000	0.0000
6	0.9756	1.3884	216.7658	1083.8292
7	0.7611	1.7749	137.6399	688.1994
8	0.7861	1.3610	161.9265	809.6325
9	0.1281	1.9654	0.0000	0.0000
10	0.4504	1.6323	58.9501	294.7507

Portfolio volatility is 500.0000 million, so the risk budget binds. Gross expected annual PnL is 819.3519 million, projected execution cost is 44.6080 million, and the skill-uncertainty charge is 127.0318 million. The objective value is consequently 647.7120 million. Total GMV is 5208.8098 million, implying annual traded notional of 52088.0982 million at $10\times$ turnover. PMs 5 and 9 receive zero allocation: their expected returns do not compensate for their incremental contribution to portfolio risk at the optimum.